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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

p.random.seed(42)
n_rows = 200

mock_data = {
    'Patient_ID': range(1001, 1001 + n_rows),
    'Age': np.random.choice([25, 45, 65, 75, 85], n_rows),
    'Disease': np.random.choice(['Flu', 'COVID-19', 'H1N1', 'SARS-CoV-2'], n_rows),
    'Admission_Date': pd.date_range(start='2020-01-01', end='2020-12-31', freq='D', n_rows)

df = pd.DataFrame(mock_data)

print("--- TASK 1: Data Cleaning ---")
print(f"Missing values before cleaning:\n{df.isnull().sum()}")

df['Age'] = df['Age'].fillna(df['Age'].mode()[0])

df = df.dropna(subset=['Disease'])

print(f"Missing values after cleaning:\n{df.isnull().sum()}")

print("--- TASK 2: Patient Breakdown ---")

bins = [0, 18, 35, 55, 75, 100]
labels = ['0-17', '18-34', '35-54', '55-75', '75-100']
df['Age_Group'] = pd.cut(df['Age'], bins=bins, labels=labels)

breakdown = df.groupby(['Disease', 'Age_Group']).size().reset_index()
print("Patient Count Matrix (Disease vs Age Group):")
print(breakdown, "\n")

print("--- TASK 3: Peak Admission Periods ---")

df['Month'] = df['Admission_Date'].dt.month
df['Day_of_Week'] = df['Admission_Date'].dt.dayofweek

peak_month = df['Month'].value_counts().sort_index().idxmax()
peak_day = df['Day_of_Week'].value_counts().sort_index().idxmax()

print(f"Peak Admission Month: {peak_month}")
print(f"Peak Admission Day of the Week: {peak_day}")

print("--- TASK 4: Generating Visualizations ---")
plt.figure(figsize=(14, 10))
plt.subplot(2, 2, 1)

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lt.subplot(2, 2, 1)
ns.countplot(data=df, x='Disease', hue=
lt.title('Patient Count by Disease')
lt.xticks(rotation=45)

lt.subplot(2, 2, 2)
ns.countplot(data=df, x='Age_Group', hue=
lt.title('Patient Count by Age Group')

lt.subplot(2, 2, 3)
f['Month_Year'] = df['Admission_Date'].c
rend_data = df.groupby('Month_Year').si
rend_data.plot(kind='line', marker='o',
lt.title('Monthly Admission Trends')
lt.ylabel('Number of Patients')

lt.subplot(2, 2, 4)
ns.heatmap(breakdown, annot=True, cmap=
lt.title('Disease Frequency by Age Group')

lt.tight_layout()
lt.show()
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--- TASK 1: Data Cleaning ---
Missing values before cleaning:
Patient_ID      0
Age             34
Disease         15
Admission_Date  0
dtype: int64

```

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Missing values after cleaning:
Patient_ID      0
Age             0
Disease         0
Admission_Date  0
dtype: int64

```

```

--- TASK 2: Patient Breakdown ---
Patient Count Matrix (Disease vs Age Group):
Age_Group    0-17  18-34  35-54  55-74  75+
Disease
Covid-19      9     20     9      7     5
Diabetes      7     20     7      6     7
Flu           5     39     5      7     6
Hypertension  6      8     4      4     4

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--- TASK 3: Peak Admission Periods ---
Peak Admission Month: January
Peak Admission Day of the Week: Wednesday

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--- TASK 4: Generating Visualizations ---

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